An Overview of Geospatial Web Services and Standards

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What are Geospatial Web Services?

- A set of technologies that allow geographic data and processing capabilities to be distributed and accessed over the Internet.
- They enable **interoperability** by defining standard protocols for communication between different software systems (e.g., a desktop GIS and a web server).
- The client (web browser, mobile app, desktop GIS) does not need to store the data or the processing logic.
- They form the core of modern, distributed Web GIS Architectures.



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Core Geospatial Standards: OGC and REST

- **Open Geospatial Consortium (OGC):**
  - The primary international standards organization for the geospatial domain.
  - Defines specifications for services that return either **images (maps)** or **data (features)**.
  - OGC services typically rely on XML/GML for data exchange.
- **Representational State Transfer (REST):**
  - An architectural style, often simpler than OGC, using standard HTTP methods (GET, POST).
  - Data is often exchanged using formats like **GeoJSON** or **JSON**.
  - Used by many modern commercial and open-source platforms (e.g., Esri's ArcGIS REST API, modern GeoServer APIs).



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Key OGC Service: Web Map Service (WMS)

- **Purpose:** Returns map images (rendered raster or vector data) over the web. It's for visualization only.
- **Operations (Key Requests):**
  - **GetCapabilities:** Returns an XML document describing the service's layers, supported coordinate systems, and image formats.
  - **GetMap:** The main request. Returns an image (PNG, JPEG, GIF) of a specified geographic area, layer(s), style, and size.
  - **GetFeatureInfo:** Returns attribute information for a specific point clicked on the map image.
- **Analogy:** Like requesting a screenshot of a map from the server. You get a picture, not the raw ingredients.



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Key OGC Service: Web Feature Service (WFS)

- **Purpose:** Allows clients to retrieve actual geographic feature data (vectors) and their attributes.
- **Data Format:** Features are typically returned in **Geography Markup Language (GML)**, a spatial XML format. GeoJSON is also supported by some WFS versions.
- **Key Operations:**
  - **GetCapabilities:** Lists the feature layers and allowed operations.
  - **GetFeature:** The core request. Returns raw vector features (Points, Lines, Polygons) and their attributes based on a query (e.g., bounding box, SQL-like filter).
  - **Transaction (WFS-T):** Allows for creating, deleting, and updating features on the server's database.
- **Analogy:** Like requesting the spreadsheet of coordinates and attributes, allowing for local analysis.



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Key OGC Service: Web Processing Service (WPS)

- **Purpose:** Standardizes how to publish and execute geospatial processes (analysis, geoprocessing) over the web.
- **Examples of Processes:** Buffer generation, overlay analysis, raster reclassification, and route finding.
- **The Process Chain:**
  1. **DescribeProcess:** Client finds out the inputs and outputs required for an analysis.
  2. **Execute:** Client sends the input data and parameters to the server.
  3. Server performs the complex analysis.
  4. Server returns the result (map, feature, or text) to the client.
- It moves the heavy computation load from the client to the server.



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The Server Component: GeoServer and MapServer

- An essential software layer that implements the OGC standards and serves the data.
- **GeoServer:**
  - Written in Java, a full-featured, robust server.
  - Excellent support for OGC standards (WMS, WFS, WPS).
  - Easy configuration and management via a web interface.
  - Integrates seamlessly with PostGIS.
- **MapServer:**
  - A CGI program, known for being fast and lightweight.
  - Excellent for serving map images (WMS) efficiently.
  - Configuration is primarily file-based (mapfiles).
- Both are completely open-source and widely used.



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Summary and Key Takeaways

- Geospatial Web Services are essential for building distributed, interoperable GIS applications.
- **WMS** is for **visualization** (returns map images).
- **WFS** is for **data access** and **editing** (returns raw features).
- **WPS** is for **analysis and geoprocessing** (performs complex tasks on the server).
- Servers like **GeoServer** and **MapServer** implement these standards, providing the connection between your spatial database (e.g., PostGIS) and the client application (e.g., Leaflet).
- Modern RESTful APIs offer a simpler, alternative approach to data serving.



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Questions?

Thank You!



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